Not Just Smart – Now AI Has a Will

Sumit Tyagi

About Sumit

Leading AI/ ML Department @ Ksolves| Ex- Lead Data Scientist @ Paytm



Taught 10,000 data scientists

Can bench-press 100 KGs

Previously.....

Consulting @ Absolutdata - Infogain

Product @ LEAD School

SME @ UpGrad



DISCLAIMER

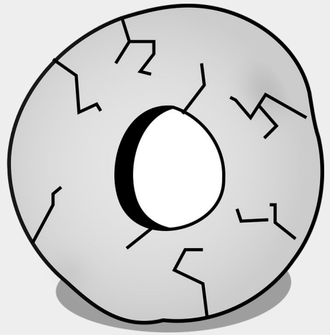


DISCLAIMER

- Crude Language
- Coarse Humour

Section 1

Humans Love Automation



Section 1

Humans use to carry things on the back



Section 1

Humans use to carry things on the back
They invented wheel





Section 1

Humans use to carry things on the back
They invented wheel





Humans use to wash clothes by hand



Section 1

Humans use to carry things on the back
They invented wheel





Humans use to wash clothes by hand
They invented washing machine





Section 1

Father: *wakes up* can you make me a cup of tea?

You:

Section 1

Father: *wakes up* can you make me a cup of tea?

You:

Makes a cup of tea

Section 1

Father: *wakes up* can you make me a cup of tea?

Makes a cup of tea

You:

Buy nescafe tea machine

Section 1

Father: *wakes up* can you make me a cup of tea?

Makes a cup of tea

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Section 1

Father: *wakes up* can you make me a cup of tea?

Makes a cup of tea

You:

Buy nescafe tea machine **automate** 👍



Section 1

Father: *wakes up* can you make me a cup of tea? *finishes tea*

Father: Can you drop me to the office I don’t feel like driving today?

Section 1

Father: *wakes up* can you make me a cup of tea? *finishes tea*

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Section 1

Father: *wakes up* can you make me a cup of tea? *finishes tea*

Father: Can you drop me to the office I don’t feel like driving today?

You:

Drop him to the office

Section 1

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You:

Drop him to the office



(you ask driver)

Section 1

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You:

Drop him to the office



(you ask driver)

automate ?

Section 1

Father: *wakes up* can you make me a cup of tea? *finishes tea*

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You:

Drop him to the office

OLA Uber (you ask driver)

automate ?



Section 1

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You:

Drop him to the office



(you ask driver)

automate ? **Self driving car (AI)**

Section 1

Remember,

- If you can write the solution of a task **on a paper,** it can be automated.
- It’s called **rule-based** approach.

- If it’s difficult for you to write it down.
- You may need to **delegate** it to a **person** or **AI**.

Section 1

Remember,

- If you can write the solution of a task **on a paper,** it can be automated.
- It’s called **rule-based** approach.

Mostly Done

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Section 1

Remember,

- If you can write the solution of a task **on a paper,** it can be automated.
- It’s called **rule-based** approach.

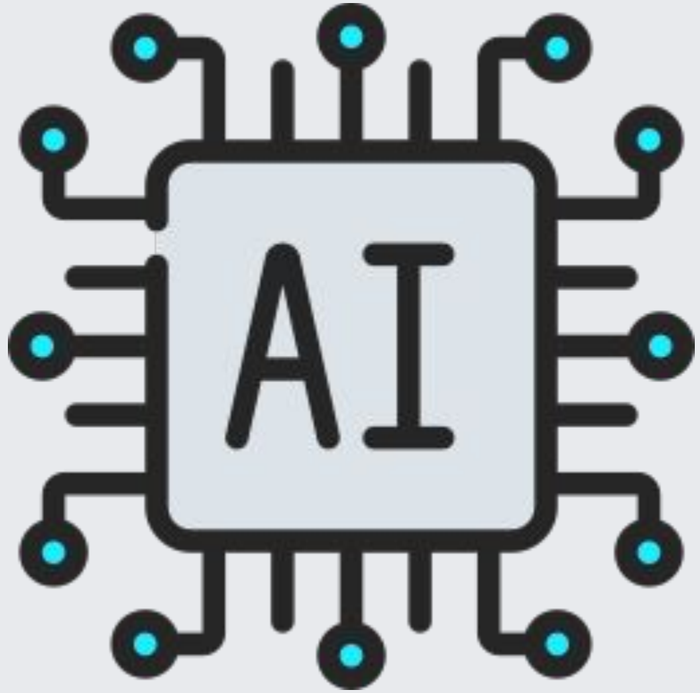
Mostly Done

- If it’s difficult for you to write it down.
- You may need to **delegate** it to a **person** or **AI**.

On it!

Section 1

- Driving a car
- Recommending Bread with Butter
- Detecting Fraud
- Create Sales plan
- Call center jobs
- Translators



Google I/O Extended

Quiz Time

Which Approach to Use?

Alphabetising a list of song titles

Rules-based

Machine Learning

Which Approach to Use?

Ranking Web Search Results

Rules-based

Machine Learning

Which Approach to Use?

Predicting Housing Prices based on Location

Rules-based

Machine Learning

Which Approach to Use?

Processing online payments

Rules-based

Machine Learning

Which Approach to Use?

Classifying an object in a photo

Rules-based

Machine Learning

Section 2

What AI used to do for you?

Section 2

What AI use to do for you?

Predict something

Section 2

What AI use to do for you?

Predict something

Predict house price in Gurgaon

No of passengers traveling in a IndiGo 6E plane

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

- Cancer cells vs normal cells
- Fraud transaction vs Genuine transaction

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

Detect an object in a image

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

Detect an object in a image

- Pneumonia in lung X-ray
- Terror camps in satellite images

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

Detect an object in a image

Section 2

What AI use to do for you?

Predict something

Classify between two or more objects

Detect an object in a image

Write a song

Create a painting

Do QnA

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Generate Something?

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But Why ?

Google I/O Extended

Section 3

What AI finds Difficult*?

previously*

Google I/O Extended

Quick Poll

How are you feeling?

Quick Poll

How are you feeling?

Quick Poll

How are you feeling?



Quick Poll

How are you feeling?





But Why ?

But Why ?

In traditional AI

We find a function maps an input to output

But Why ?

In traditional AI

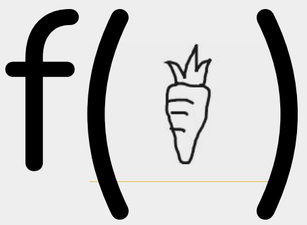
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In traditional AI

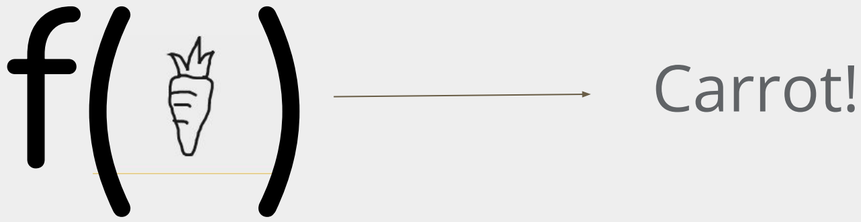
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But Why ?

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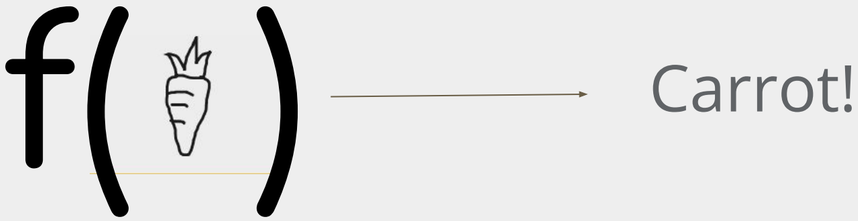
We find a function maps an input to output



But Why ?

In traditional AI

We find a function maps an input to output



So, output depends on input.

But Why ?

Say, I need to generate a text.

Q: What fruit you like?

A:

But Why ?

Say, I need to generate a text.

Q: What fruit you like?

A: I like apples

But Why ?

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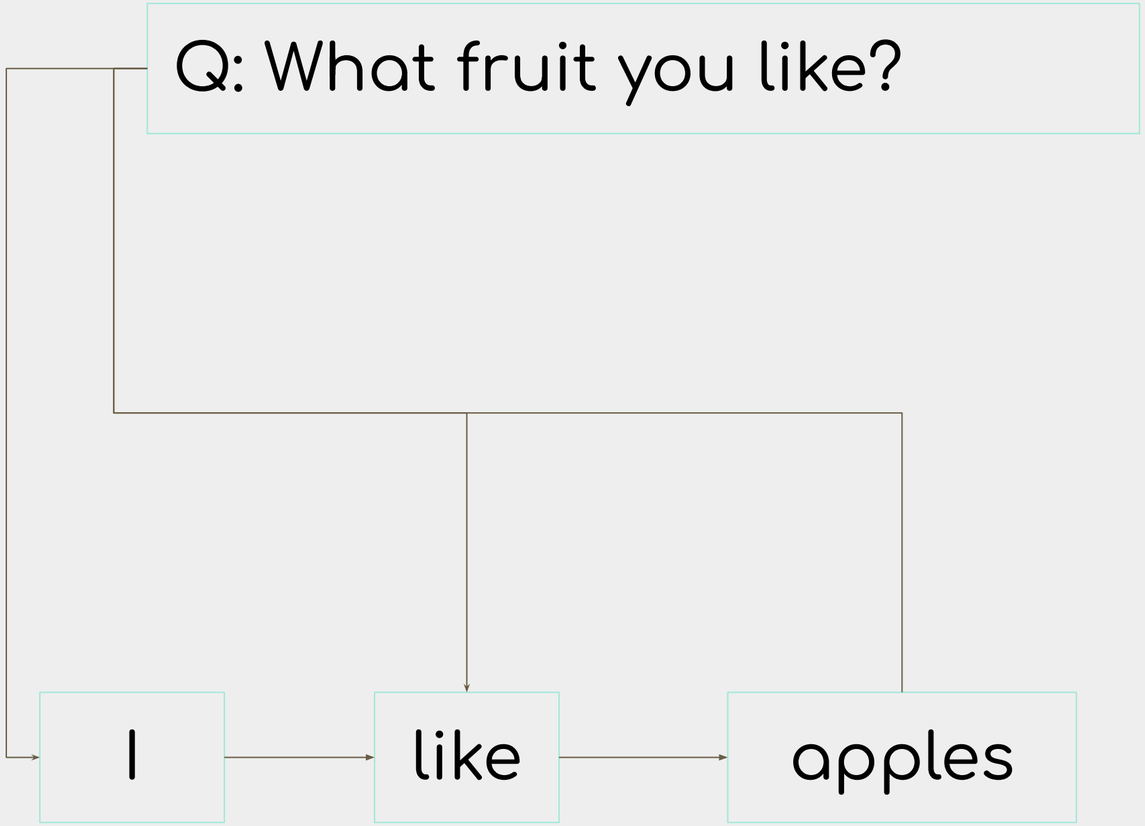
I

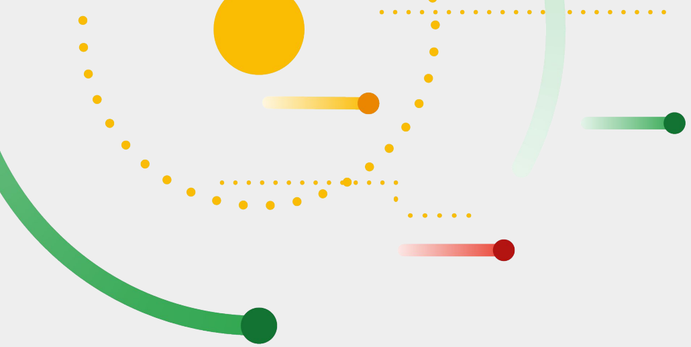
like

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But Why ?

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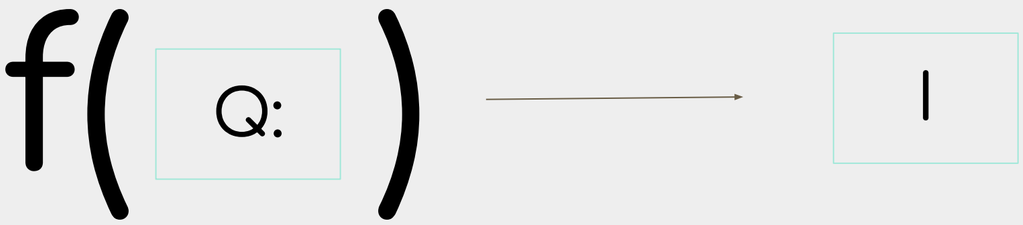


Now what?

How to Generate?

Use recurrence/ memory

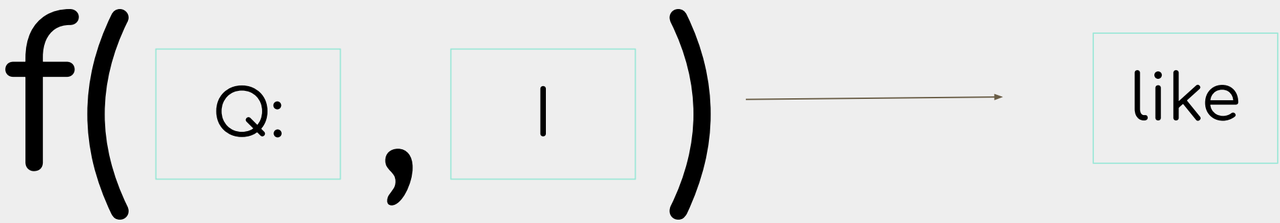
Recurrent Neural Networks & Long Short Term Memory solved it somewhat.



So, output depends on input.

Use recurrence/ memory

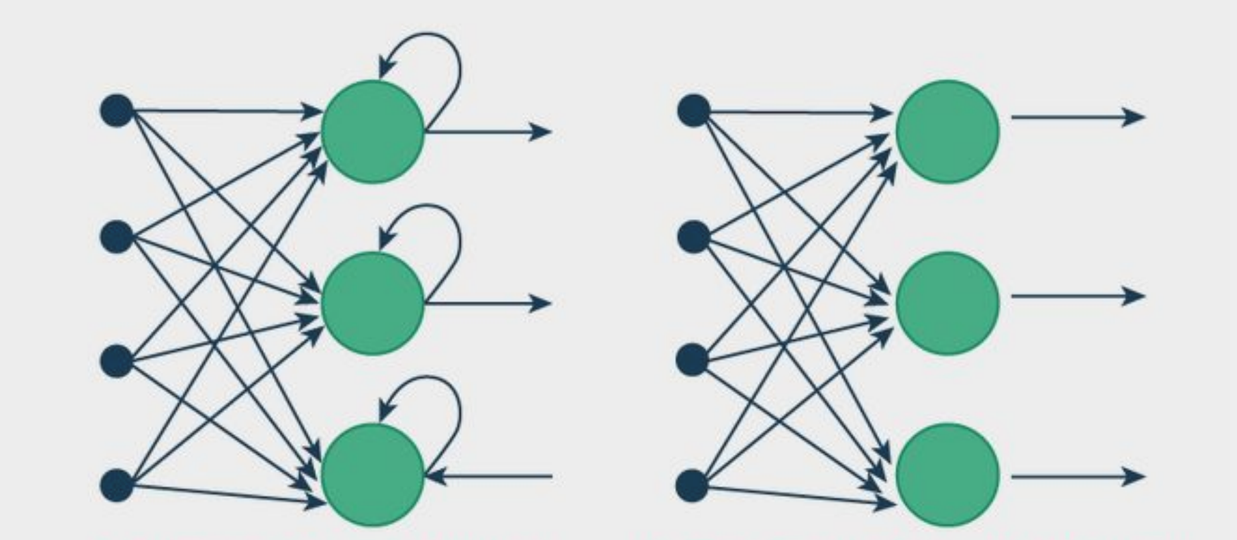
Recurrent Neural Networks & Long Short Term Memory solved it somewhat.



So, output depends on input & **Previous Output**.

Use recurrence/ memory

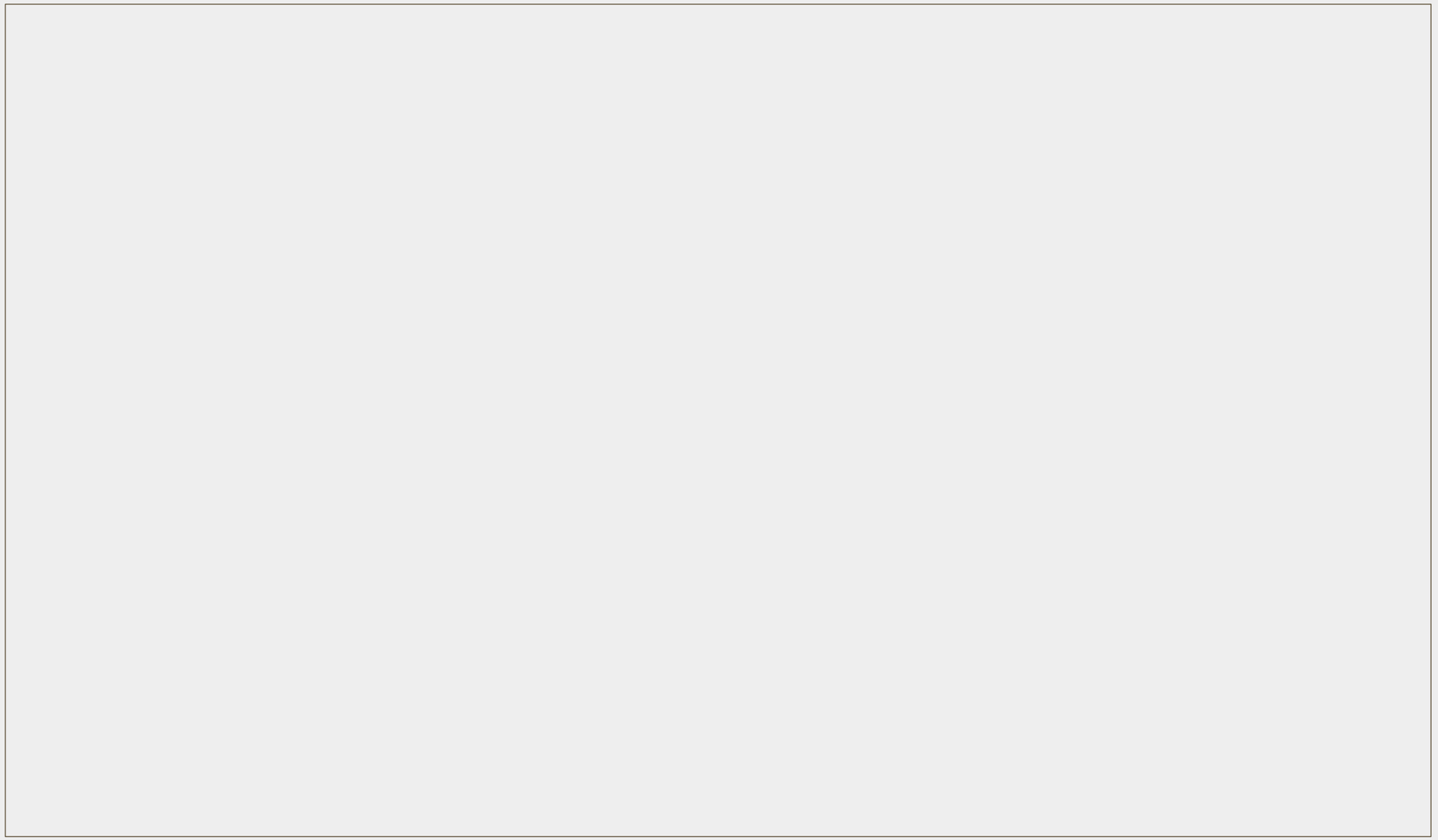
Recurrent Neural Networks & Long Short Term Memory solved it somewhat.



That’s it?

That’s it? No!

Ambiguity



That’s it? No!

Ambiguity

I want to eat apple

That’s it? No!

Ambiguity

I want to eat apple

मैं सेब खाना चाहता हूँ

That’s it? No!

Ambiguity

I want to eat apple

मैं सेब खाना चाहता हूँ

Apple iPhone is the best

That’s it? No!

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एप्पल आईफोन सबसे अच्छा है

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Sumit haridwar gya tha

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Sumit haridwar gya tha

Kahan gya tha (location)

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NER

Sumit haridwar gya tha

Kahan gya tha (location)

Kon gya tha (person)

That’s it? No!

Ambiguity	I want to eat apple	मैं सेब खाना चाहता हूँ
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NER	Sumit haridwar gya tha	
	Kahan gya tha (location)	
	Kon gya tha (person)	
Coreference		

That’s it? No!

Ambiguity	I want to eat apple	मैं सेब खाना चाहता हूँ
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Coreference	Sumit haridwar gya tha, vo kawad leke aya h	

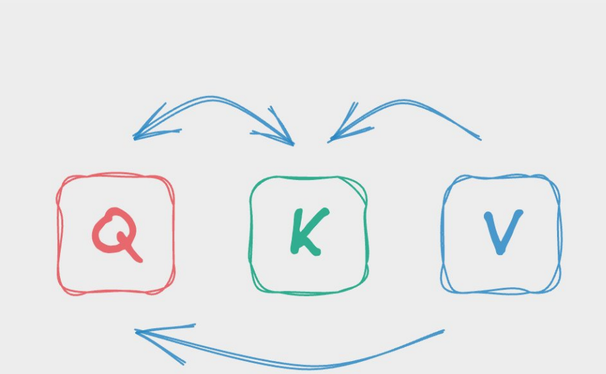
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Ambiguity	I want to eat apple	मैं सेब खाना चाहता हूँ
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Ab kya kare guru…

Section 4

Attention

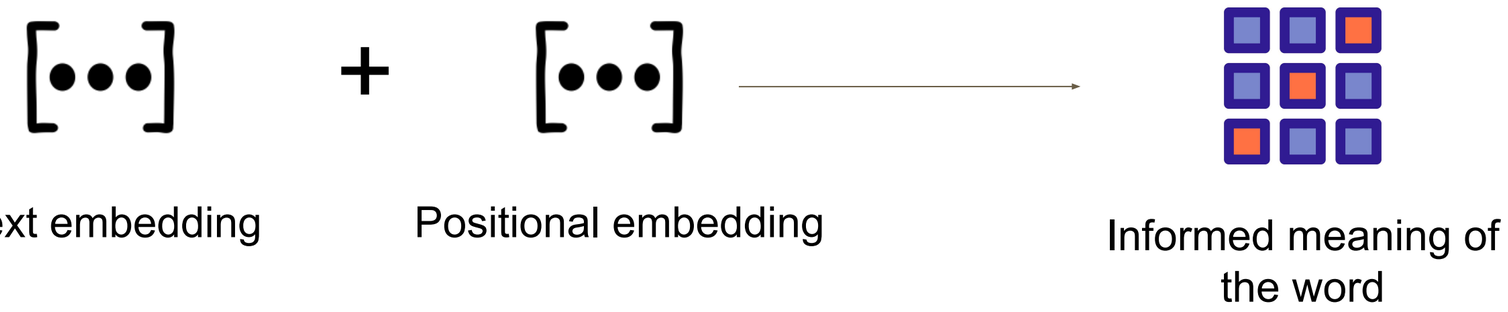


Attention is all you need

Attention says,

It’s not just important to focus on the word to understand its meaning

But also focus on the words its surrounded by.



Attention is all you need

Let’s see…

Attention is all you need



Attention is all you need

I	like	apple

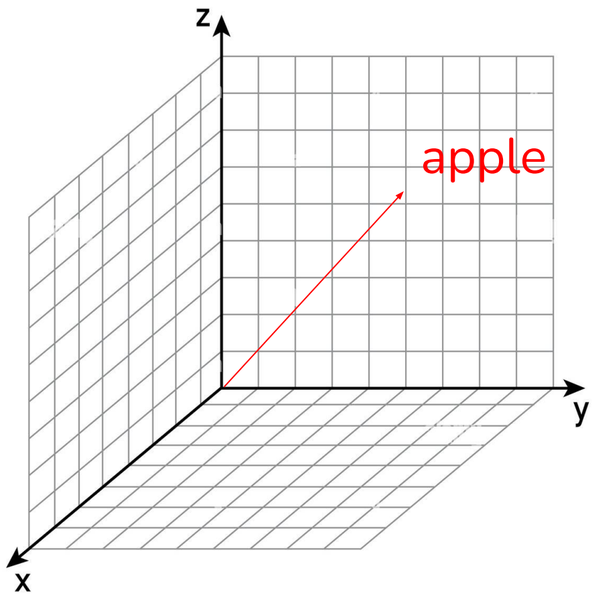
apple	iPhone	is	best

Attention is all you need



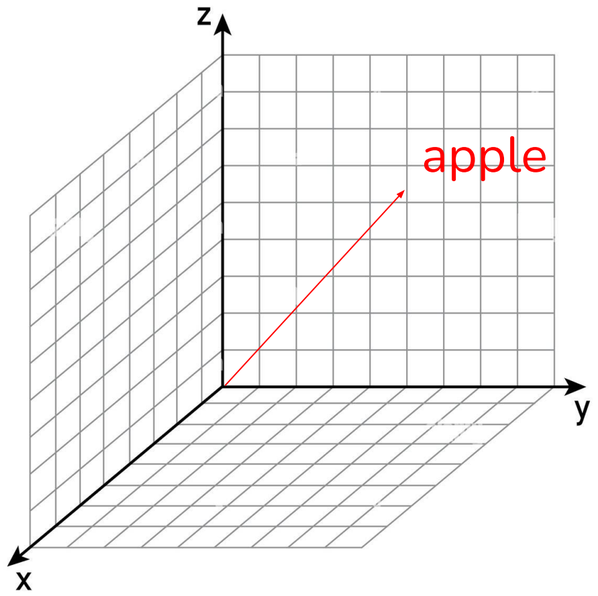
Attention is all you need

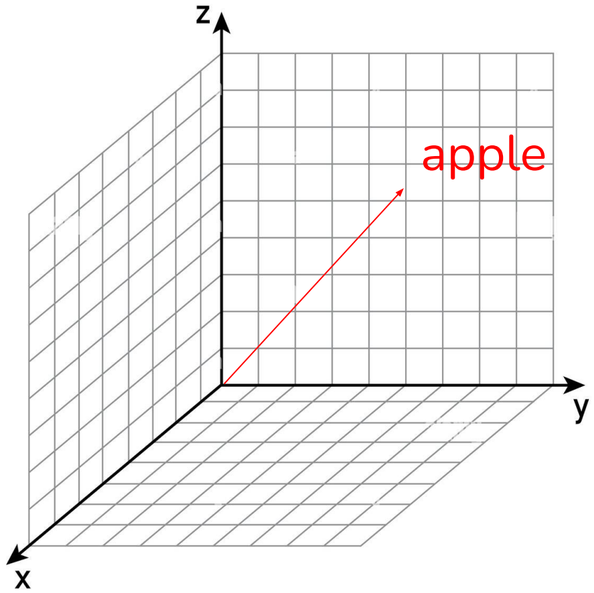
I	like	apple		apple	iPhone	is	best

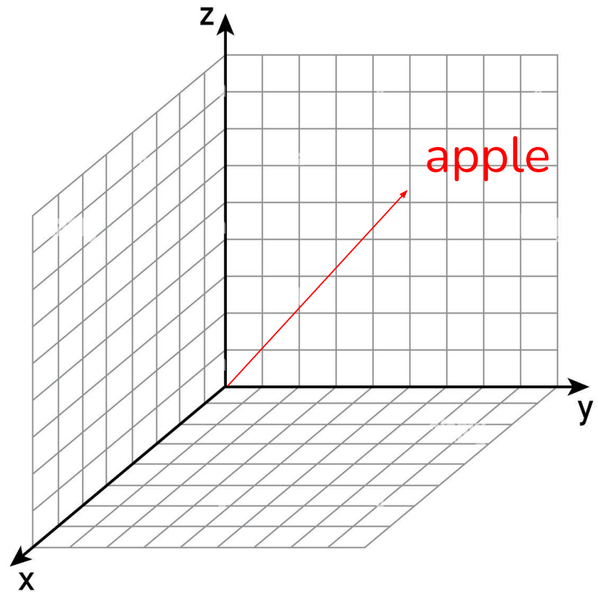


Attention is all you need

I	like	apple	apple	iPhone	is	best



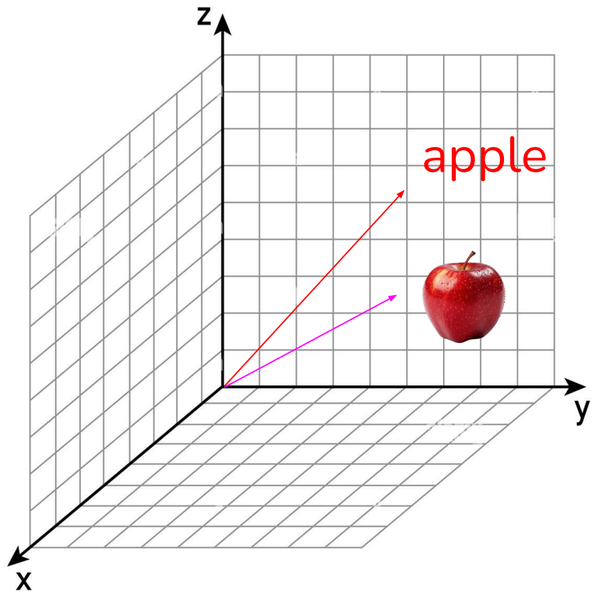


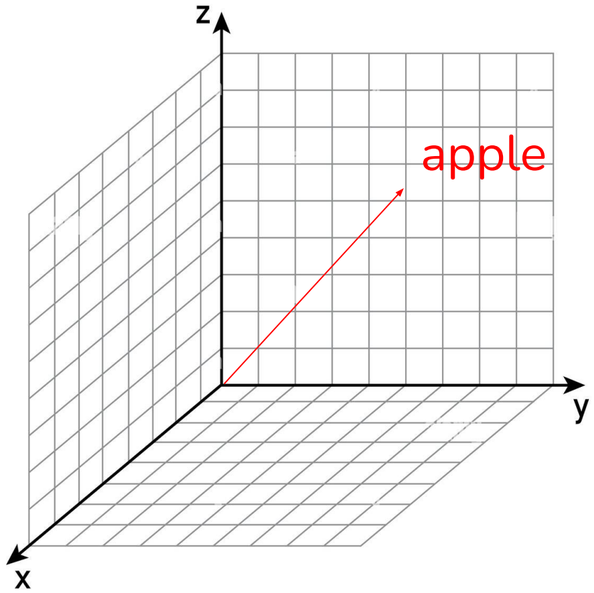


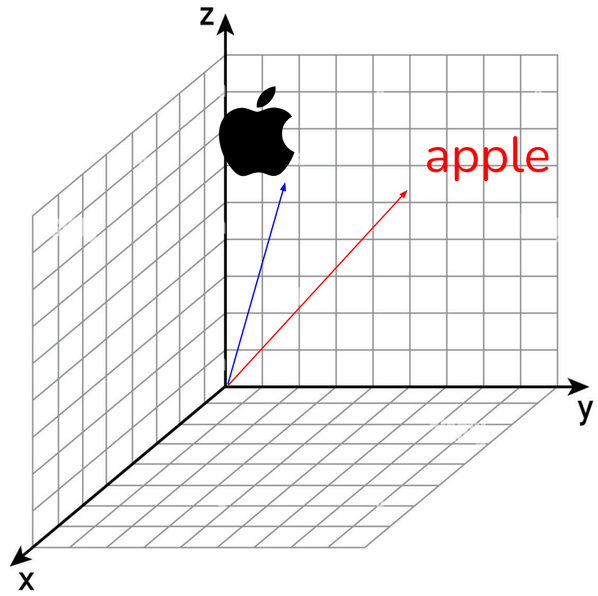
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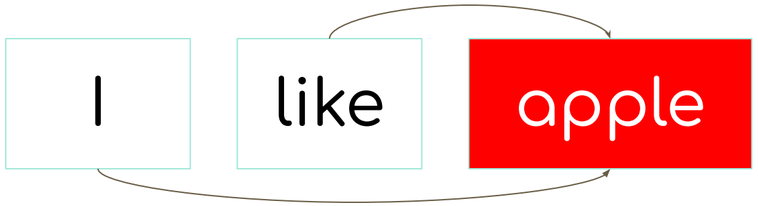
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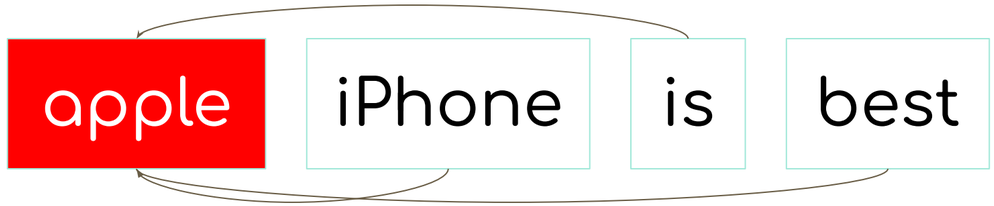


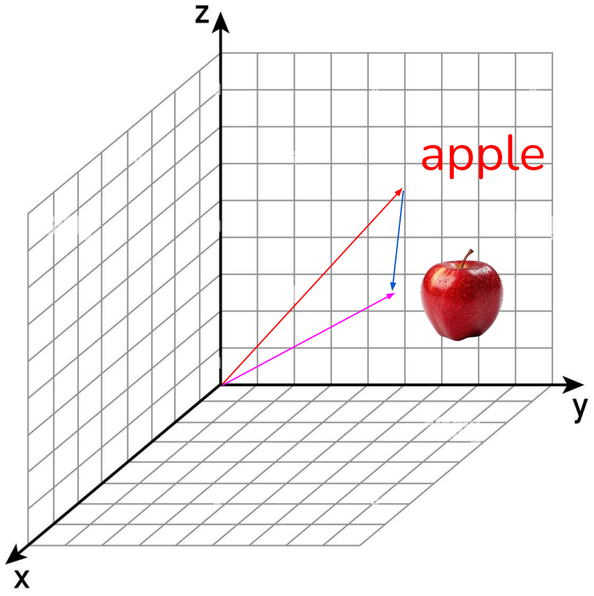


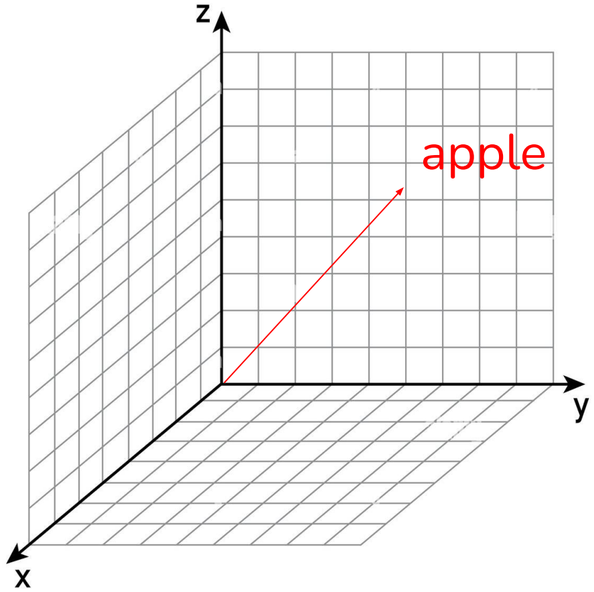


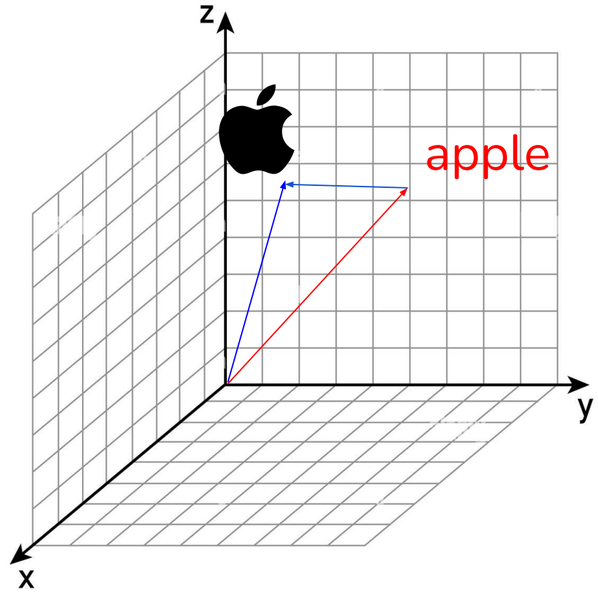
Attention is all you need



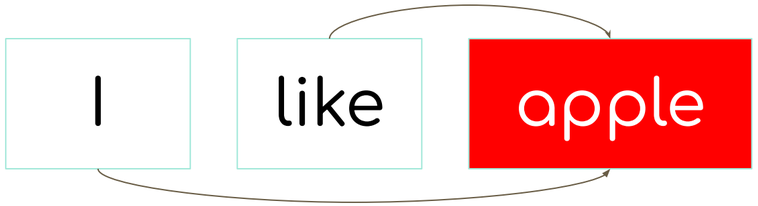


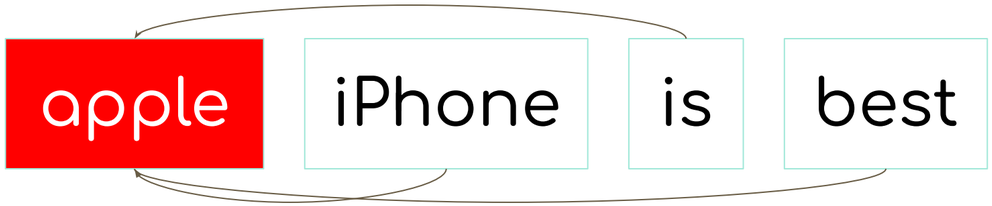






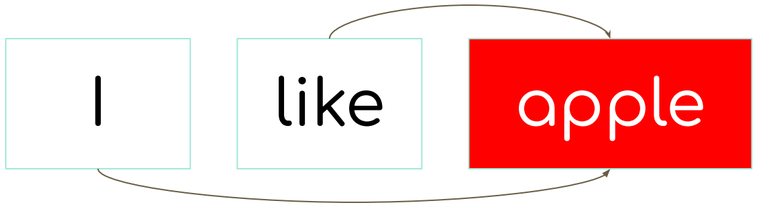
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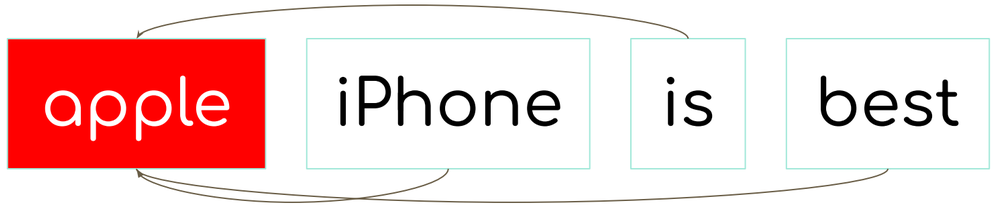




Maths: Let’s see class notes!

Attention is all you need





Let’s visualize it also


Ye to thik h guru, pr
use kaha karun?

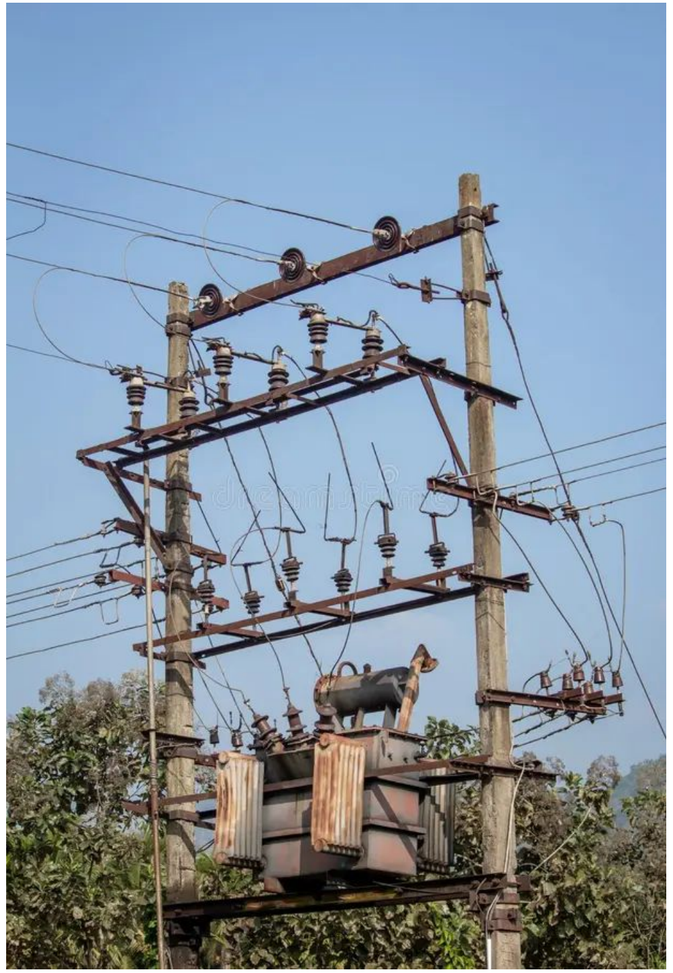
Transformers



Transformers



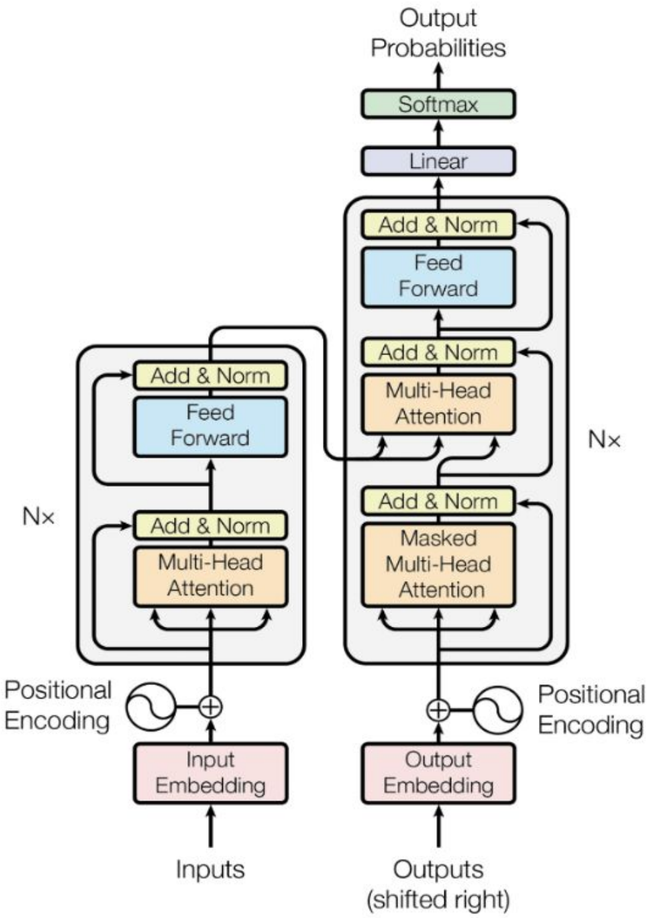
Transformers



The guy she tells you not to worry about

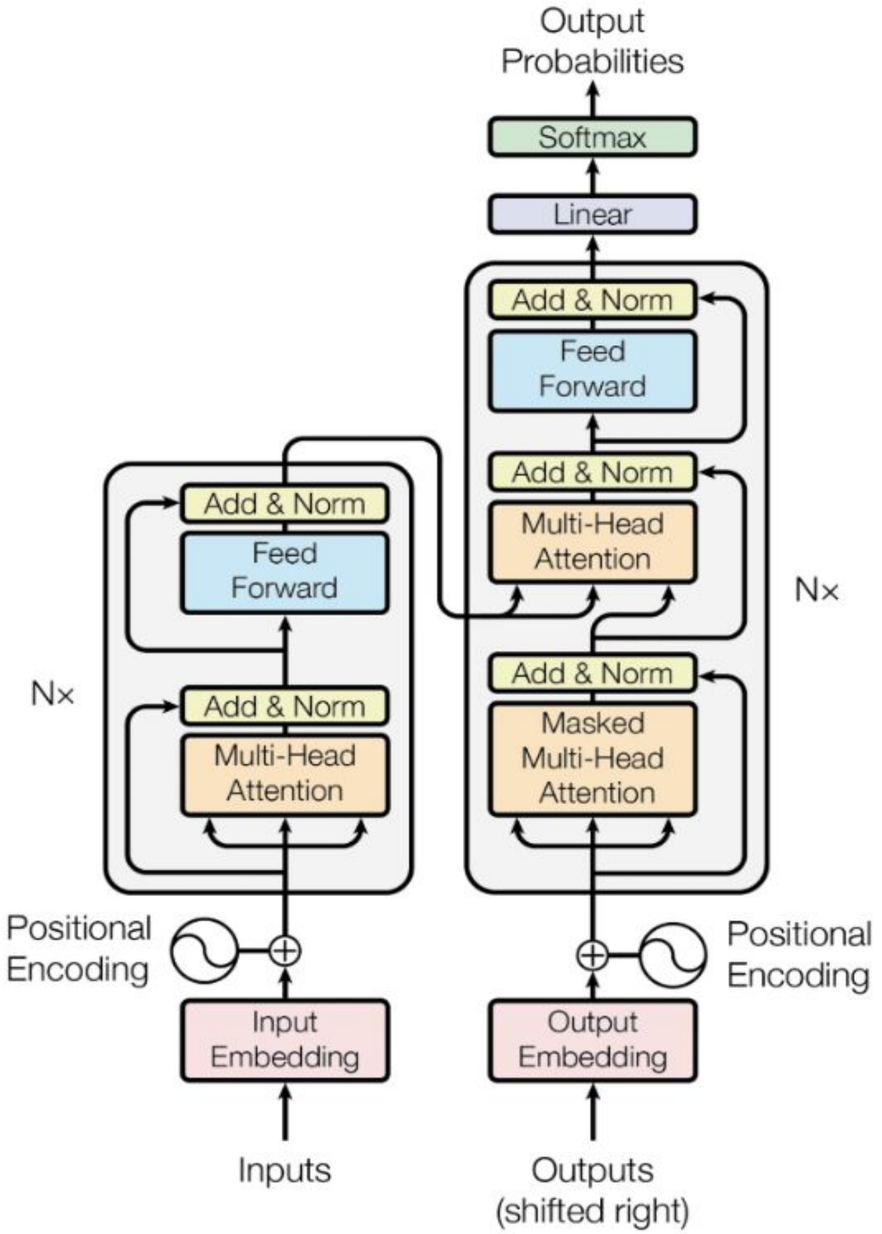


The guy you're worrying about



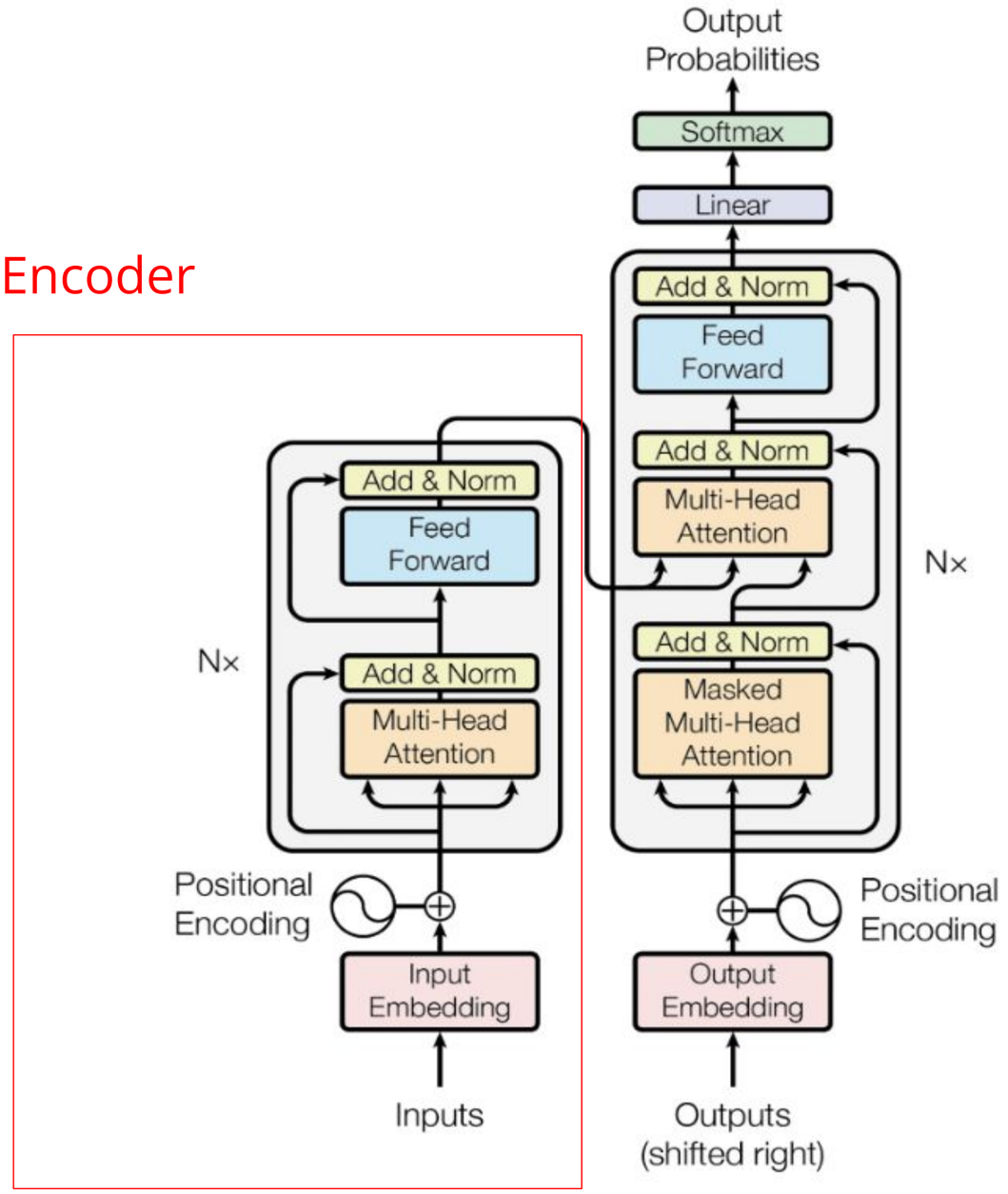
The guy you should be worry about

Transformers

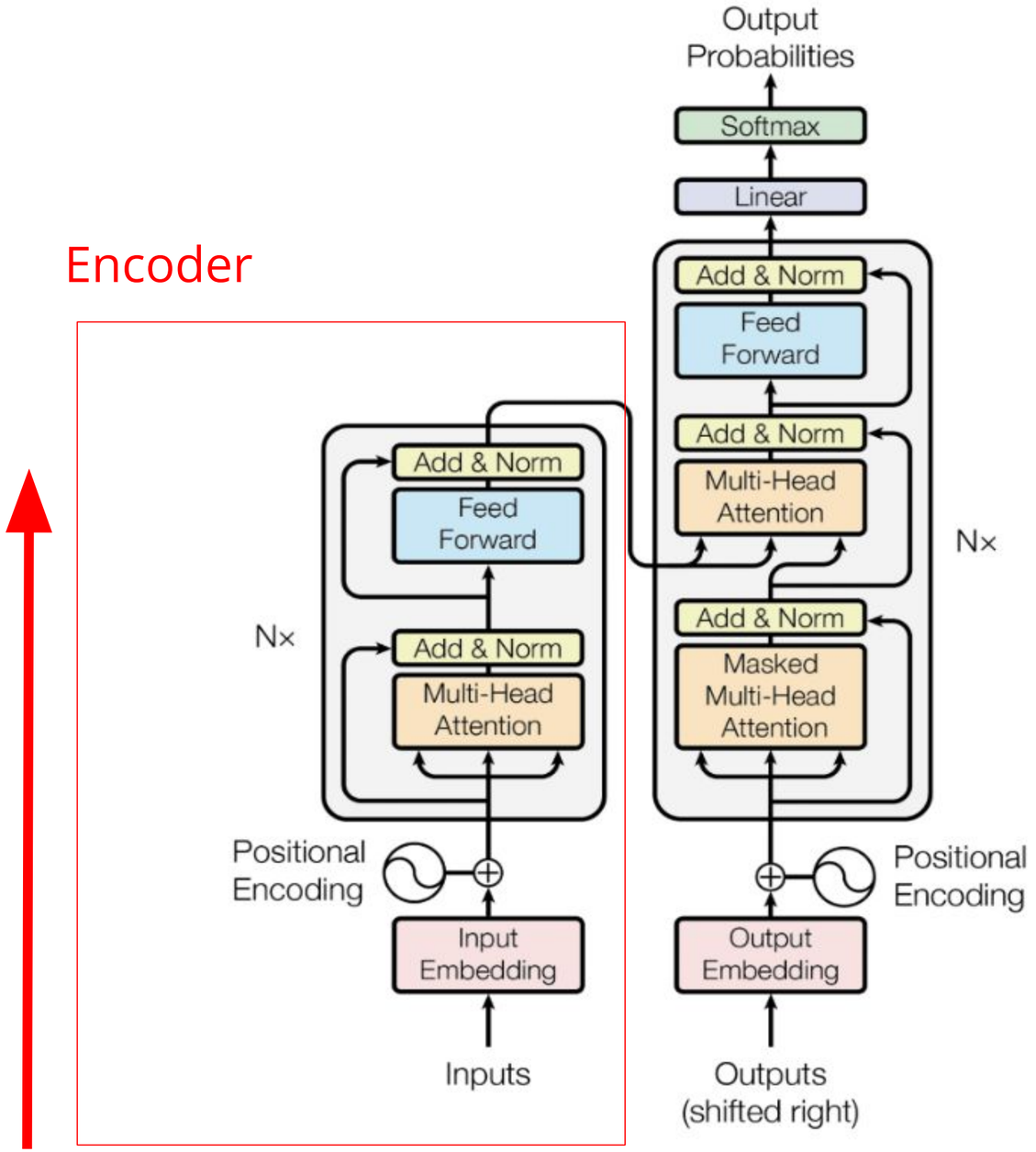


Transformers

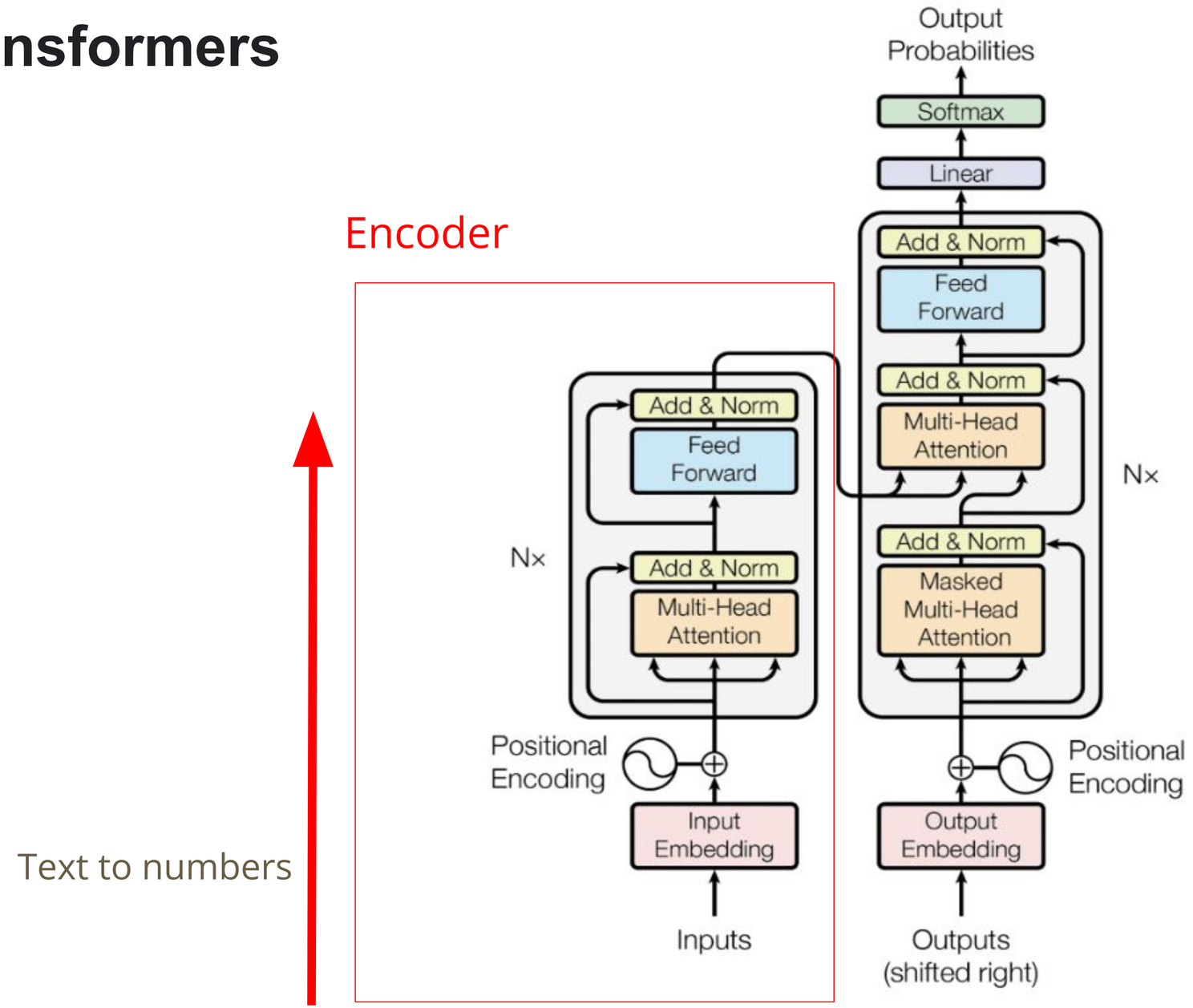
Encoder



Transformers



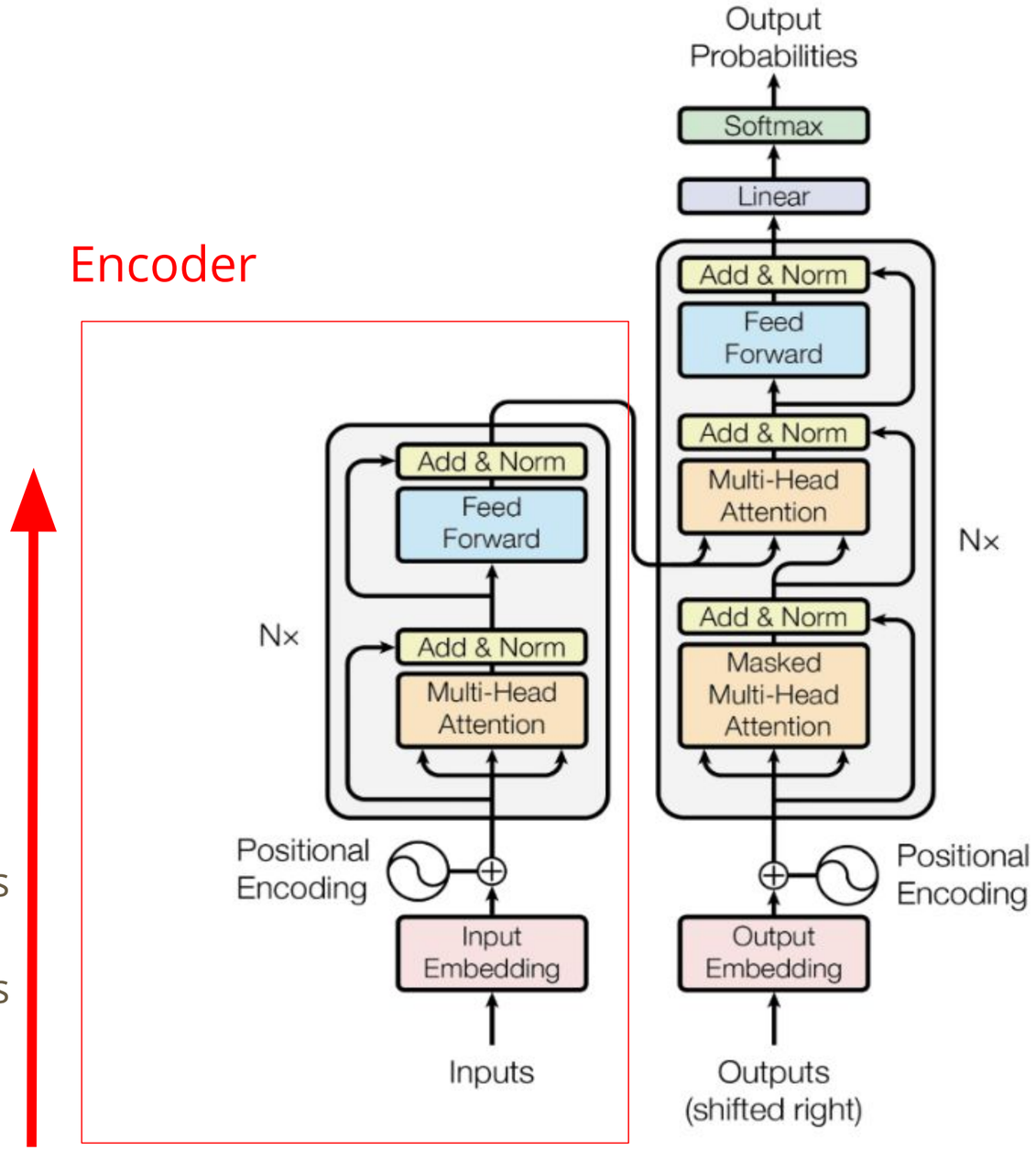
Transformers



Transformers

Position to numbers

Text to numbers



Transformers

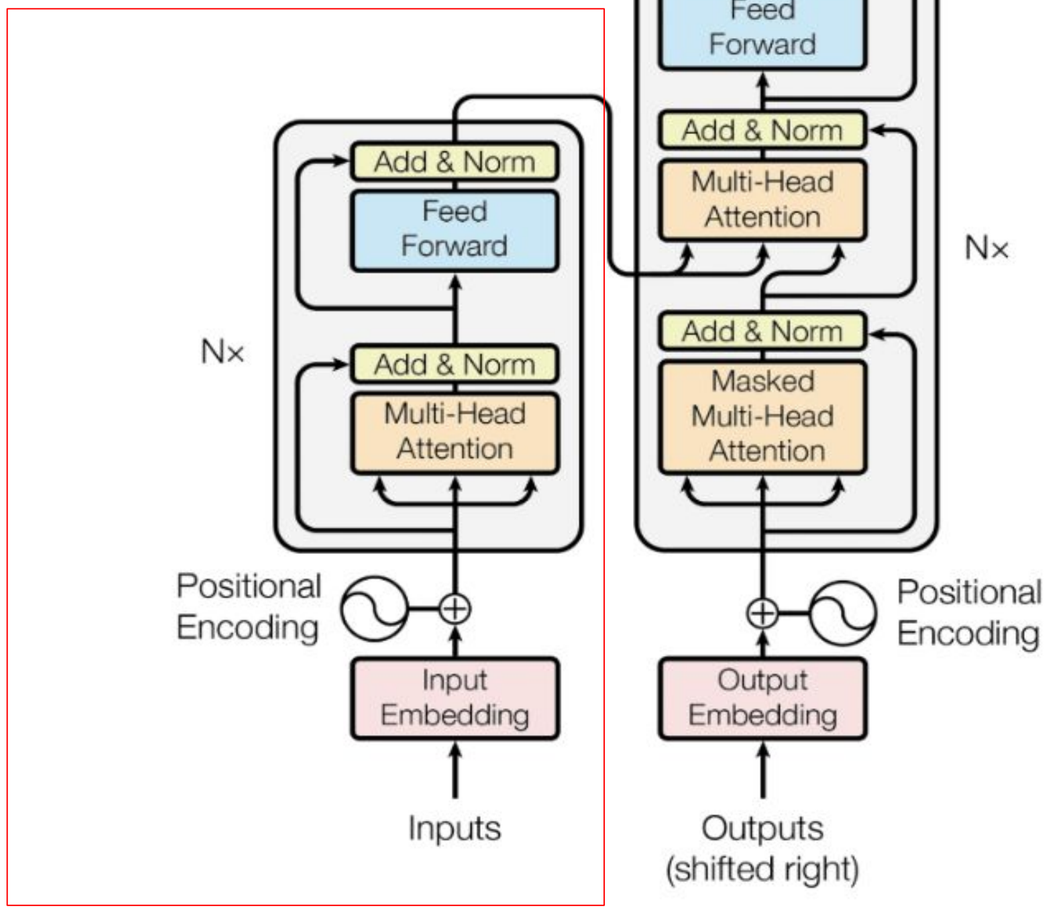
$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Position to numbers

Text to numbers

Encoder



Transformers

Sentiment, NER

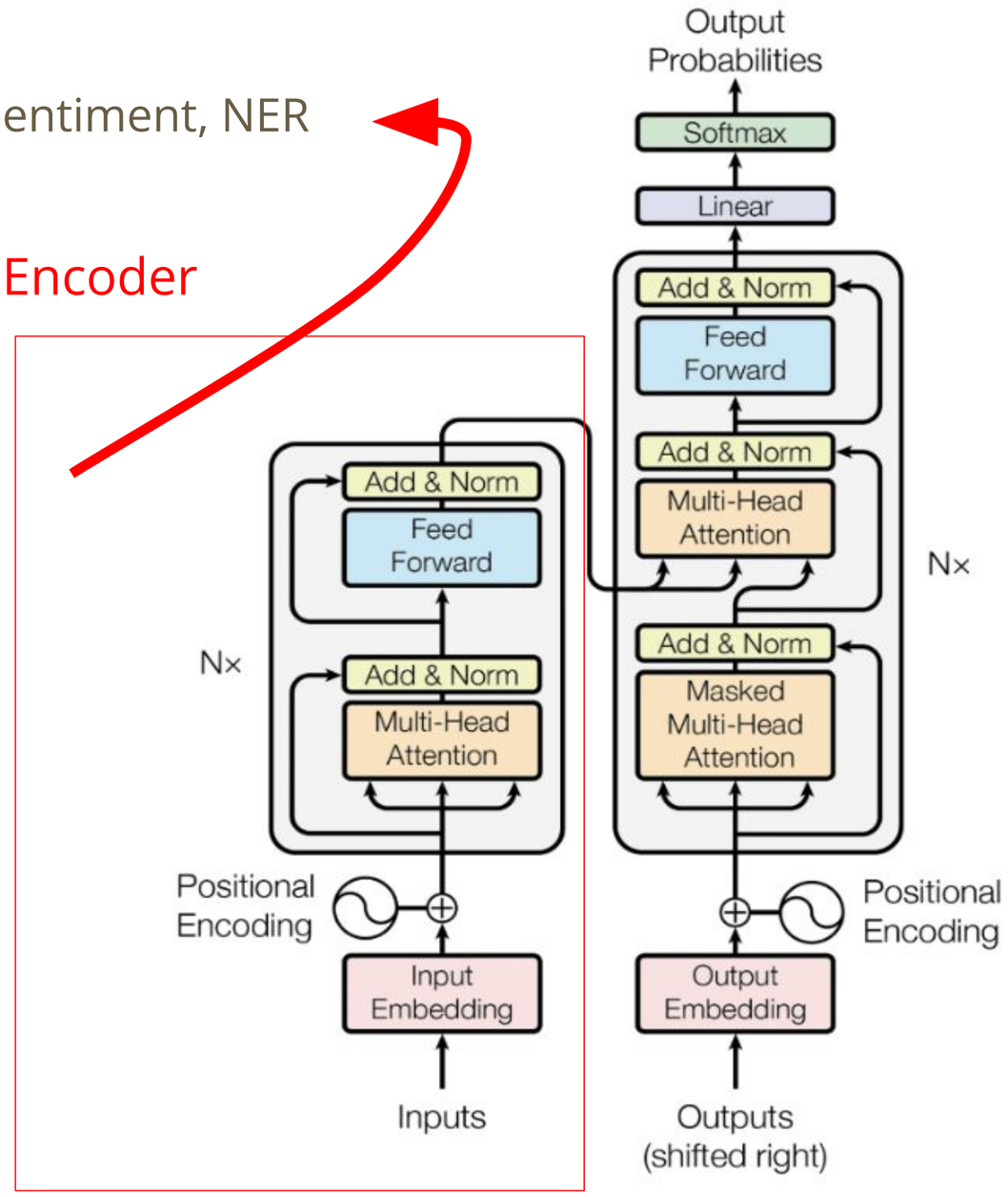
Encoder

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Position to numbers

Text to numbers



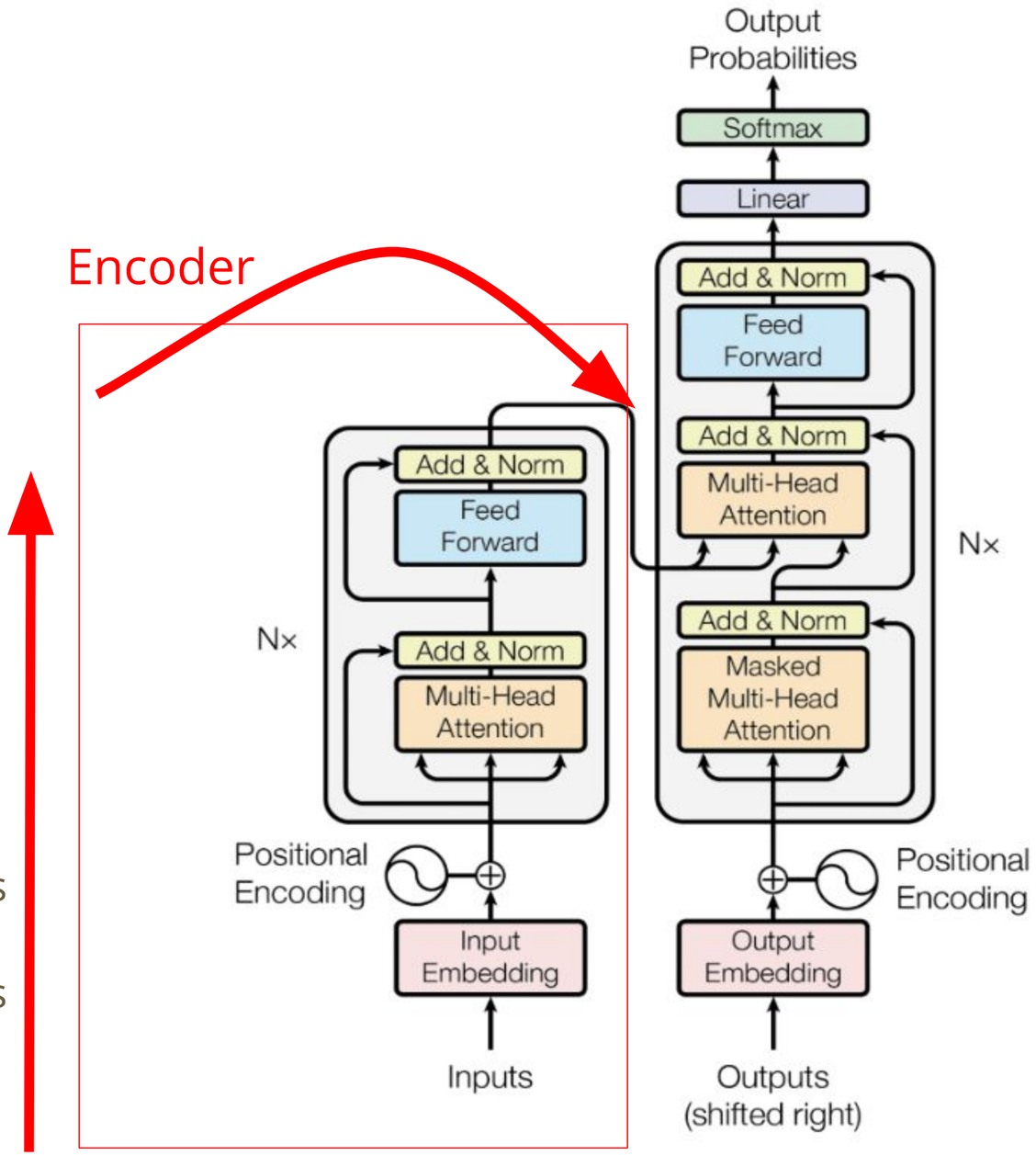
Transformers

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Position to numbers

Text to numbers



Transformers

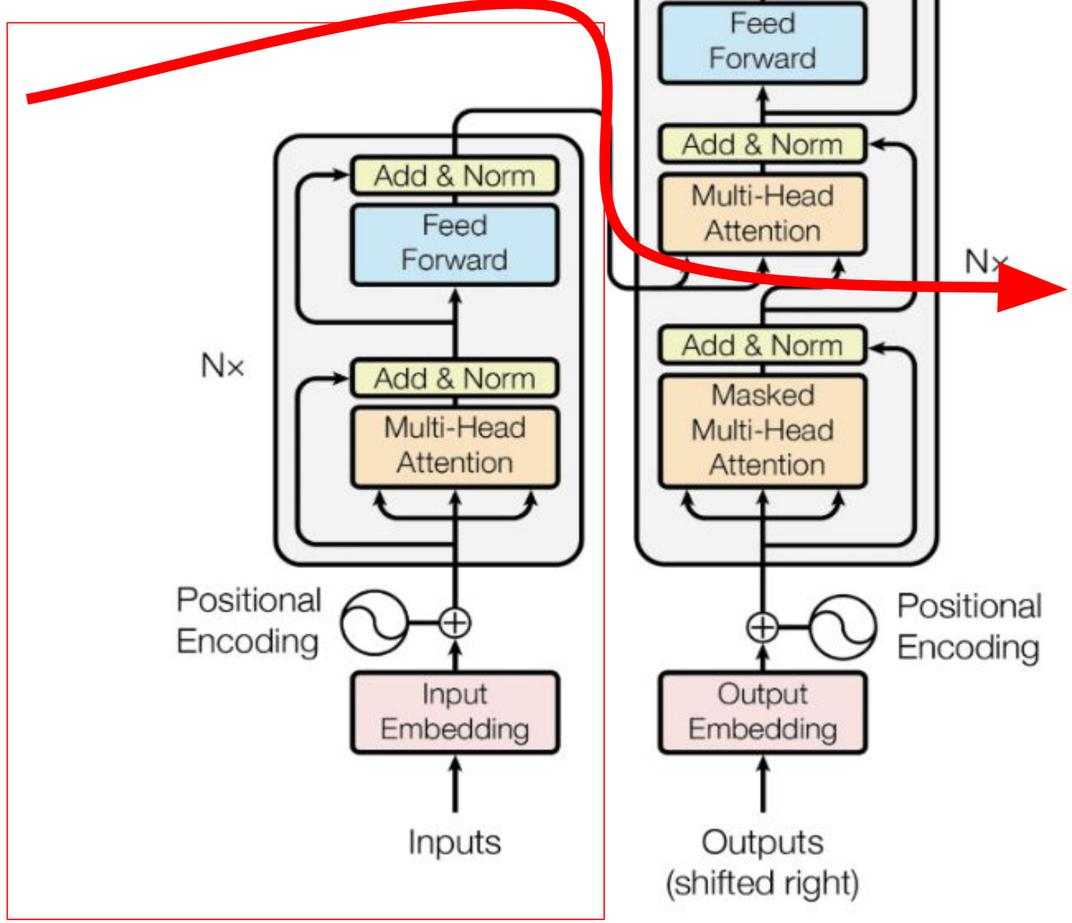
$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Position to numbers

Text to numbers

Encoder



Generate, Translate, Summarize

Position to numbers

Text to numbers

Transformers

ENCODER ONLY

aka

auto-encoding models

TASKS

- Sentence classification
- Named entity recognition
- Extractive question-answering
- Masked language modeling

EXAMPLES

BERT, RoBERTa, distilBERT

DECODER ONLY

aka

auto-regressive models

TASKS

- Text generation
- Causal language modeling

EXAMPLES

GPT-2, GPT Neo, GPT-3

ENCODER-DECODER

aka

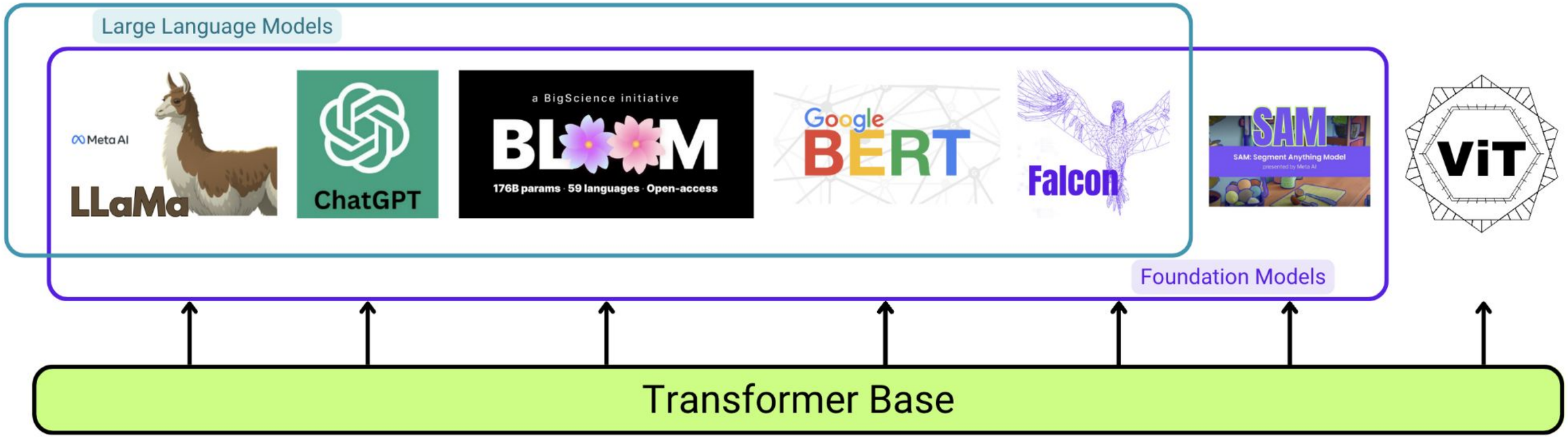
sequence-to-sequence models

TASKS

- Translation
- Summarization
- Generative question-answering

EXAMPLES

BART, T5, Marian


Large Language Models
Meta AI
LLaMa
ChatGPT
a BigScience initiative
BLOOM
176B params · 59 languages · Open-access
Google
BERT
Falcon
SAM
SAM: Segment Anything Model
presented by Meta AI
ViT
Foundation Models
Transformer Base



That’s how AI is able to do generative Tasks

Section 5

Bolti hi h karti nahi?

What about Act?



Section 5

Karegi kese?

Q: what is 1 + 2? →  → A: 3

Q: Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for $2 per egg. How much does she make every day? → 

Q: who is the latest UK PM? → 

Q: what is the prime factorization of 34324329? → 

Section 5

Karegi kese?

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Section 5

Karegi kese?

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Section 5

Karegi kese?

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Q: who is the latest UK PM? →  → Requires knowledge

Q: what is the prime factorization of 34324329? →  → Requires computation

Section 5

Reasoning?

Why Can’t Machines Learn Like Humans?

A Simple Toy Problem

Let’s begin with a basic problem to illustrate the challenge. Extract the last letter of each word and concatenate them.

Input	Output
Sumit Tyagi	ti
Bill Gates	ls
Barack Obama	?

Section 5

Reasoning?

For ML to truly mimic human intelligence, it should be able to learn from just a few examples—just like we do. However, ML struggles with this.

Why can humans grasp concepts so quickly while ML needs massive amounts of data?



You



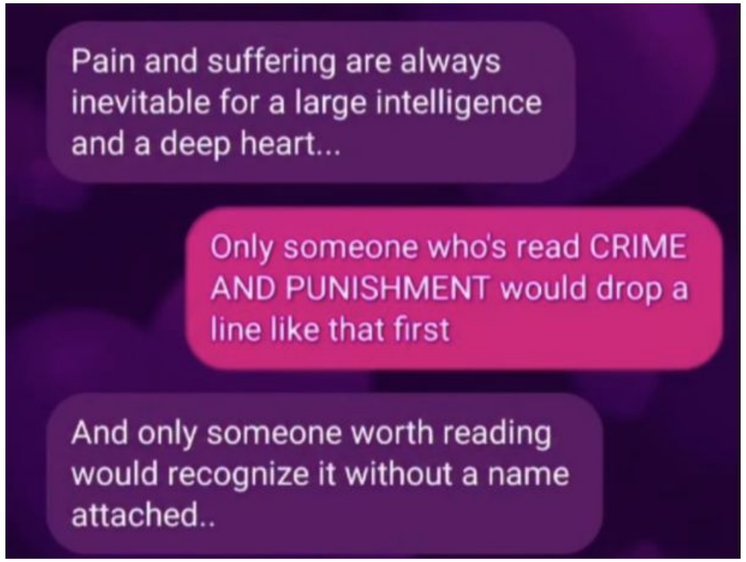
Model

Section 5

Reasoning?

For ML to truly mimic human intelligence, it should be able to learn from just a few examples—just like we do. However, ML struggles with this.

Humans don’t just memorize patterns; **we reason**, generalize, and infer based on our experiences. This allows us to learn efficiently from minimal examples.



Section 5

Reasoning?

Humans don’t just memorize patterns; **we reason**, generalize, and infer based on our experiences. This allows us to learn efficiently from minimal examples.

Can we make models reason?



Section 5

Reasoning?

Can we make models reason?

Language Models are Few-Shot Learners

Q: "Elon Musk"
A: the last letter of "Elon" is "n". the last letter of "Musk" is "k". concatenating "n", "k" leads to "nk". so the output is "nk"

Q: "Barack Obama"
A:

"ka"

Section 5

Reasoning?

Can we make models reason?

Language Models are Few-Shot Learners

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"ka"

Independent Research

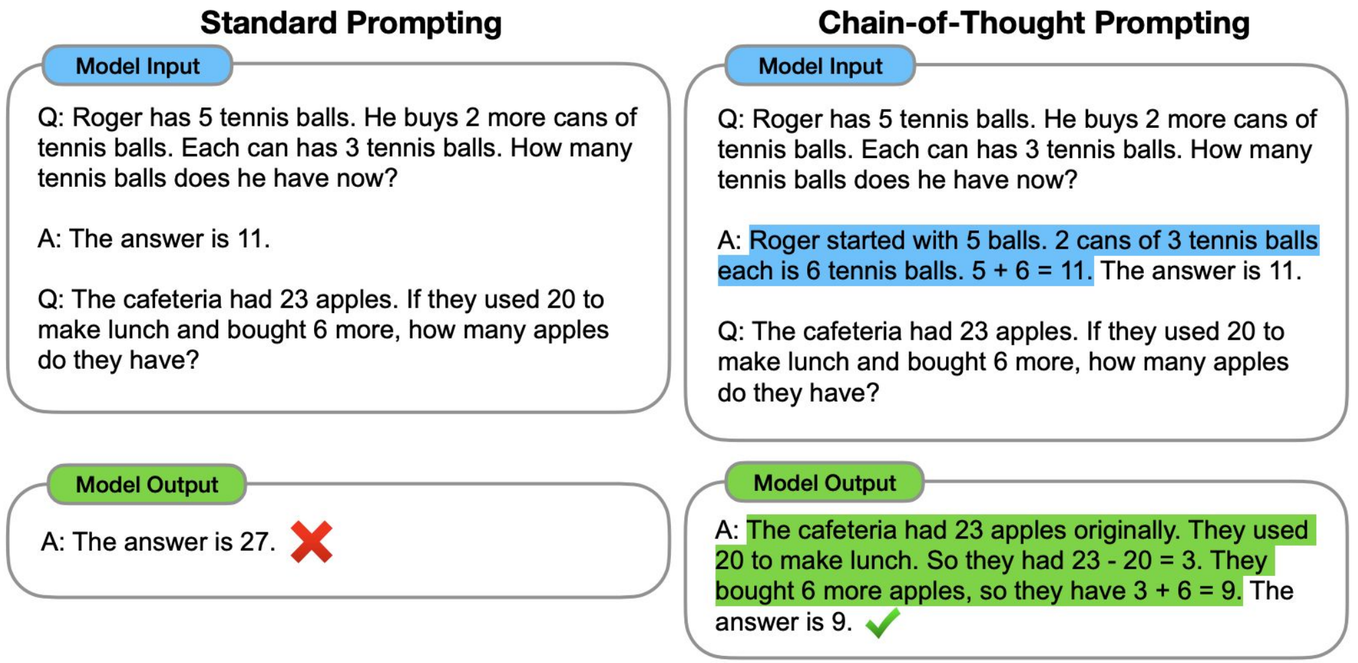
- Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems arXiv:1705.04146 [cs.AI]
- GSM8K - Training Verifiers to Solve Math Word Problems. OpenAI built a much larger math word problem dataset arXiv:2110.14168 [cs.LG]
- Show Your Work: Scratchpads for Intermediate Computation with Language Models arXiv:2112.00114 [cs.LG]

Section 5

Reasoning?

Can we make models reason?

Chain-of-Thought Prompting

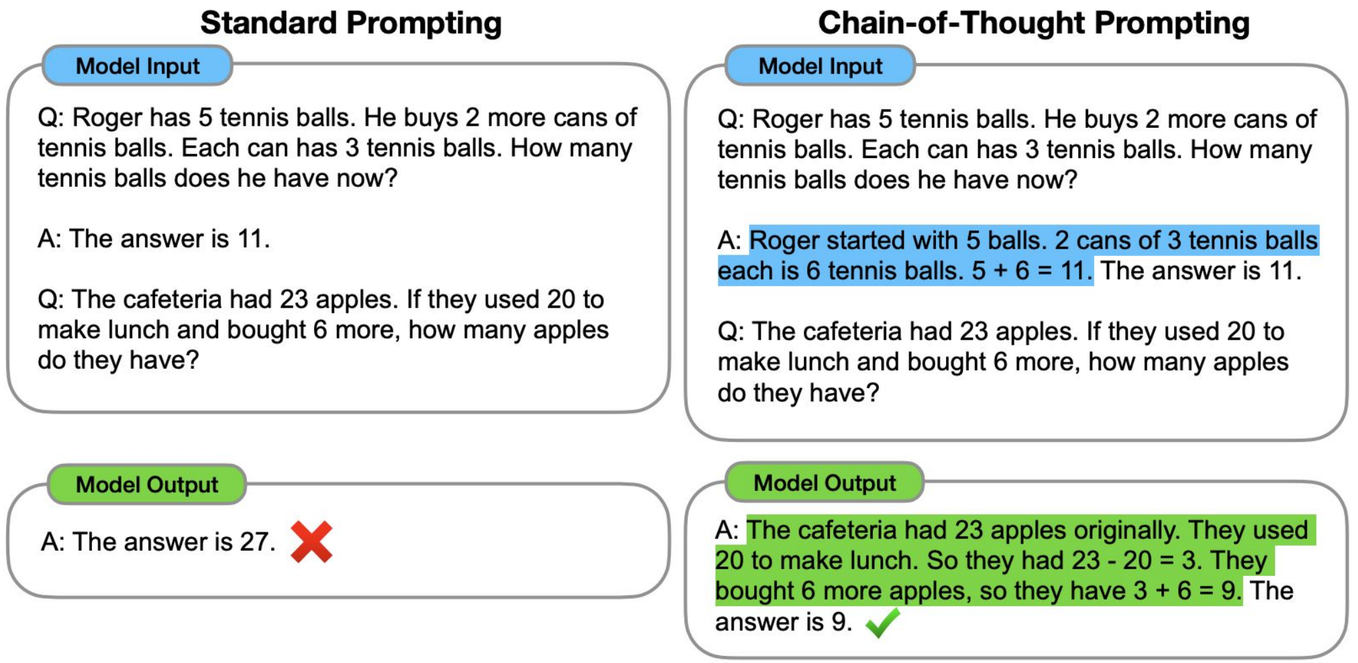


Section 5

Reasoning?

Can we make models reason?

Chain-of-Thought Prompting



Section 5

Reasoning?

Can we make models reason?

Least to Most Prompting

Question: Elsa has 3 apples. Anna has 2 more apples than Elsa. How many apples do they have together?
Decomposition: Let's break down this problem:

1. How many apples does Anna have?

2. How many apples do Elsa and Anna have together?

Response:

1. Anna has 2 more apples than Elsa. So Anna has 2 + 3 = 5 apples.

2. Anna has 5 apples. Elsa and Anna have 3 + 5 = 8 apples together.

The answer is 8.

Section 5

Reasoning?

Can we make models reason? Is it possible to trigger step-by-step reasoning without using demonstration examples?

Large Language Models are Zero-Shot Reasoners

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Section 5

Reasoning?

Can we make models reason?
still zero-shot but can do much better work?

LLMs are Analogical Reasoners

“**Do you know a related problem?** We can likely imagine a problem absolutely new, unlike and unrelated to any formerly solved problem, but if such a problem could exist, it would be insoluble” -

Section 5

Karegi kese?

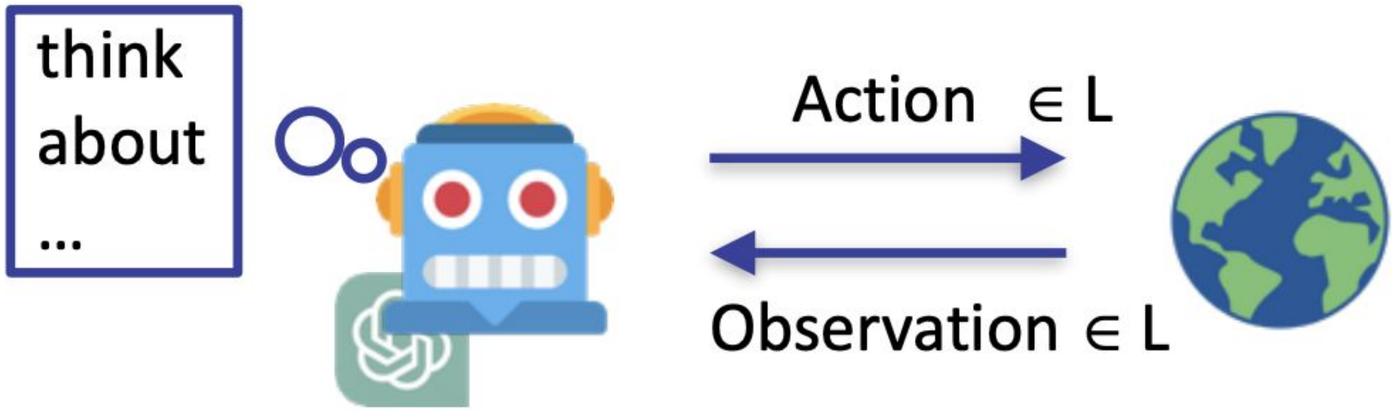
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Q: who is the latest UK PM? →  → Requires knowledge

Q: what is the prime factorization of 34324329? →  → Requires computation

Karegi kese?



Section 5

Karegi kese? People come up with Solutions



- Code augmentation

Question: In Fibonacci sequence, it follows the rule that each number is equal to the sum of the preceding two numbers. Assuming the first two numbers are 0 and 1, what is the 50th number in Fibonacci sequence?

The first number is 0, the second number is 1, therefore, the third number is 0+1=1. The fourth number is 1+1=2. The fifth number is 1+2=3. The sixth number is 2+3=5. The seventh number is 3+5=8. The eighth number is 5+8=13.
….. (Skip 1000 tokens)
The 50th number is 32,432,268,459.

CoT

32,432,268,459 ✗

```
length_of_fibonacci_sequence = 50
fibonacci_sequence = np.zeros(length_of_)
fibonacci_sequence[0] = 0
fibonacci_sequence[1] = 1
For i in range(3, length_of_fibonacci_sequence):
    fibonacci_sequence[i] = fibonacci_sequence[i-1] +
    fibonacci_sequence[i-2]
ans = fibonacci_sequence[-1]
```

PoT

python

12,586,269,025 ✓

Section 5

Karegi kese? People come up with Solutions



- Retrieval-augmented generation
  - Answer knowledge-intensive questions with
    - Extra corpora
    - A retriever (e.g., BM25, DPR, etc.)
  - What if there's no corpora? (e.g. who's the latest PM?)

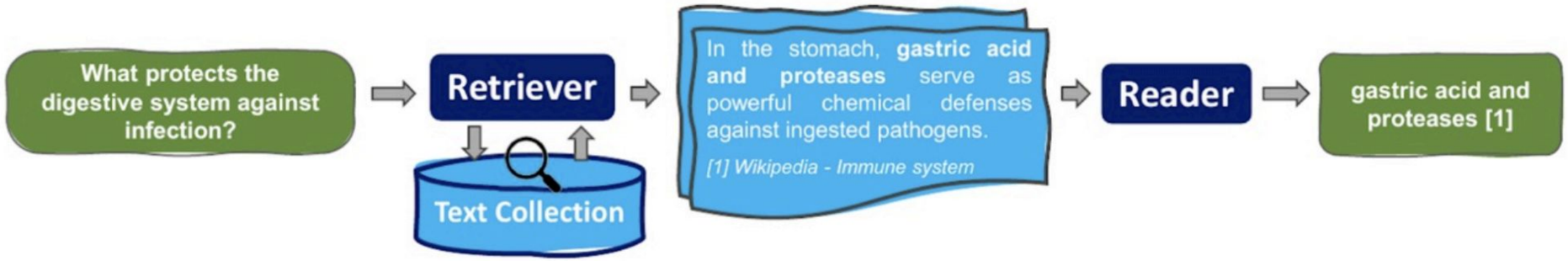


Image: http://ai.stanford.edu/blog/retrieval-based-NLP/

Section 5

Karegi kese? People come up with Solutions



- Tool use
  - Special tokens to invoke tool calls for
    - Search engine, calculator, etc.
    - Task-specific models (translation)
    - APIs
  - Unnatural format requires task/tool-specific fine-tuning
  - Multiple tool calls?

A weather task:
how hot will it get in NYC today? |*weather* lookup region=NYC |*result* precipitation chance: 10, high temp: 20c, low-temp: 12c |*output* today's high will be 20C

TALM: Tool Augmented Language Models.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Toolformer: Language Models Can Teach Themselves to Use Tools

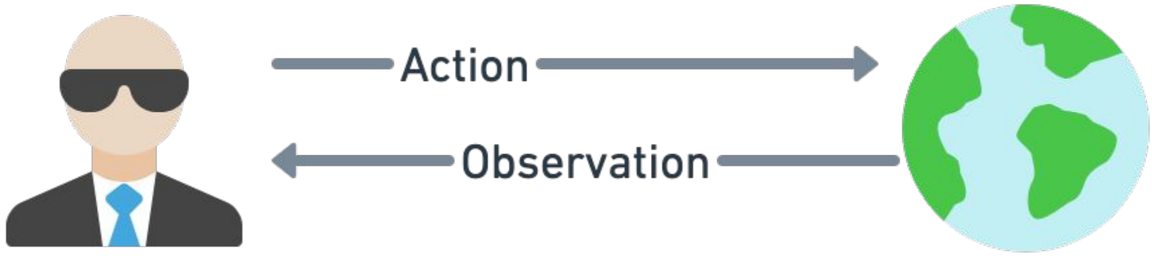
Section 5

Agent!



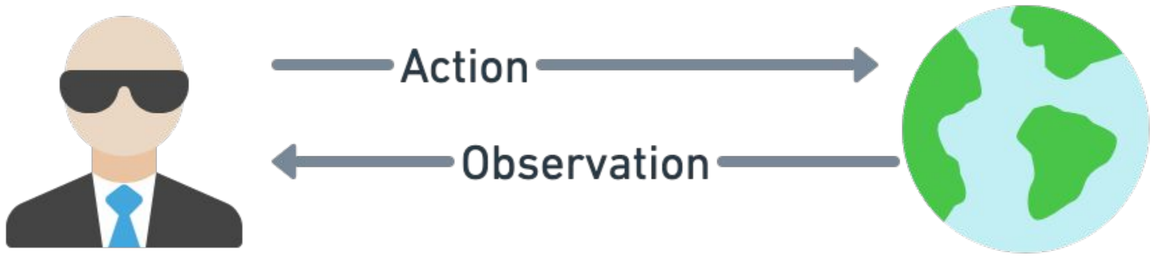
Section 5

Agent!



Section 5

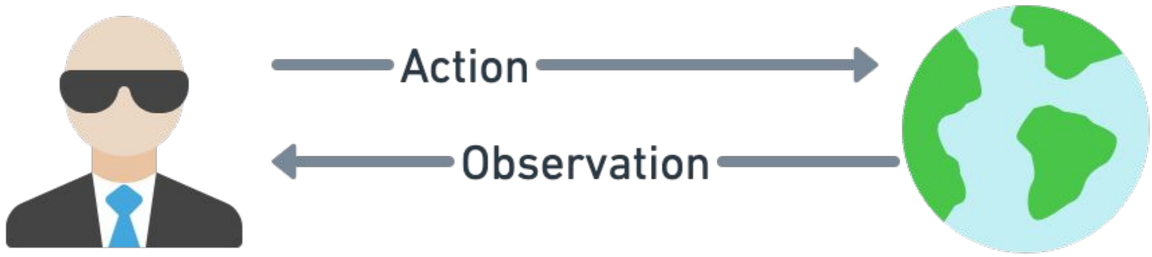
Agent!



- An “intelligent” system that interacts with some “environment”.

Section 5

Agent!



- An “intelligent” system that interacts with some “environment”.
  - Physical environments: robot, autonomous car, …
  - Digital environments: DQN for Atari, Siri, AlphaGo, …
  - Humans as environments: chatbot
- Define “agent” by defining “intelligent” and “environment”
  - It changes over time!
  - Exercise question: how would you define “intelligent”?

What's "LLM agent"?

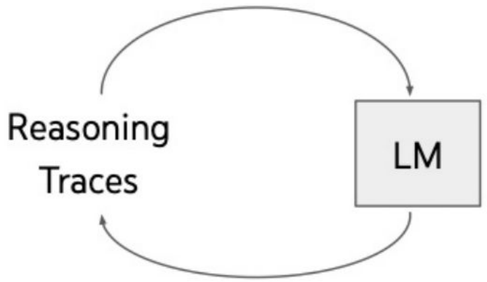


- Level 1: Text agent
  - Uses text action and observation
  - Examples: ELIZA, LSTM-DQN
- Level 2: LLM agent
  - Uses LLM to act
  - Examples: SayCan, Language Planner
- Level 3: Reasoning agent
  - Uses LLM to reason to act
  - Examples: ReAct, AutoGPT

Section 5

Bolegi ya Karegi?
Reason or Act?

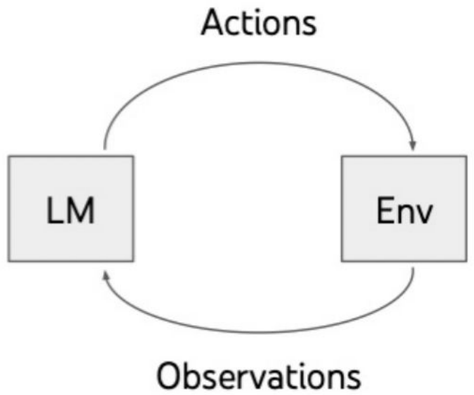
CoT



Flexible and general to
augment test-time compute

Lack of external knowledge and tools

RAG/Retrieval/Code/Tool use

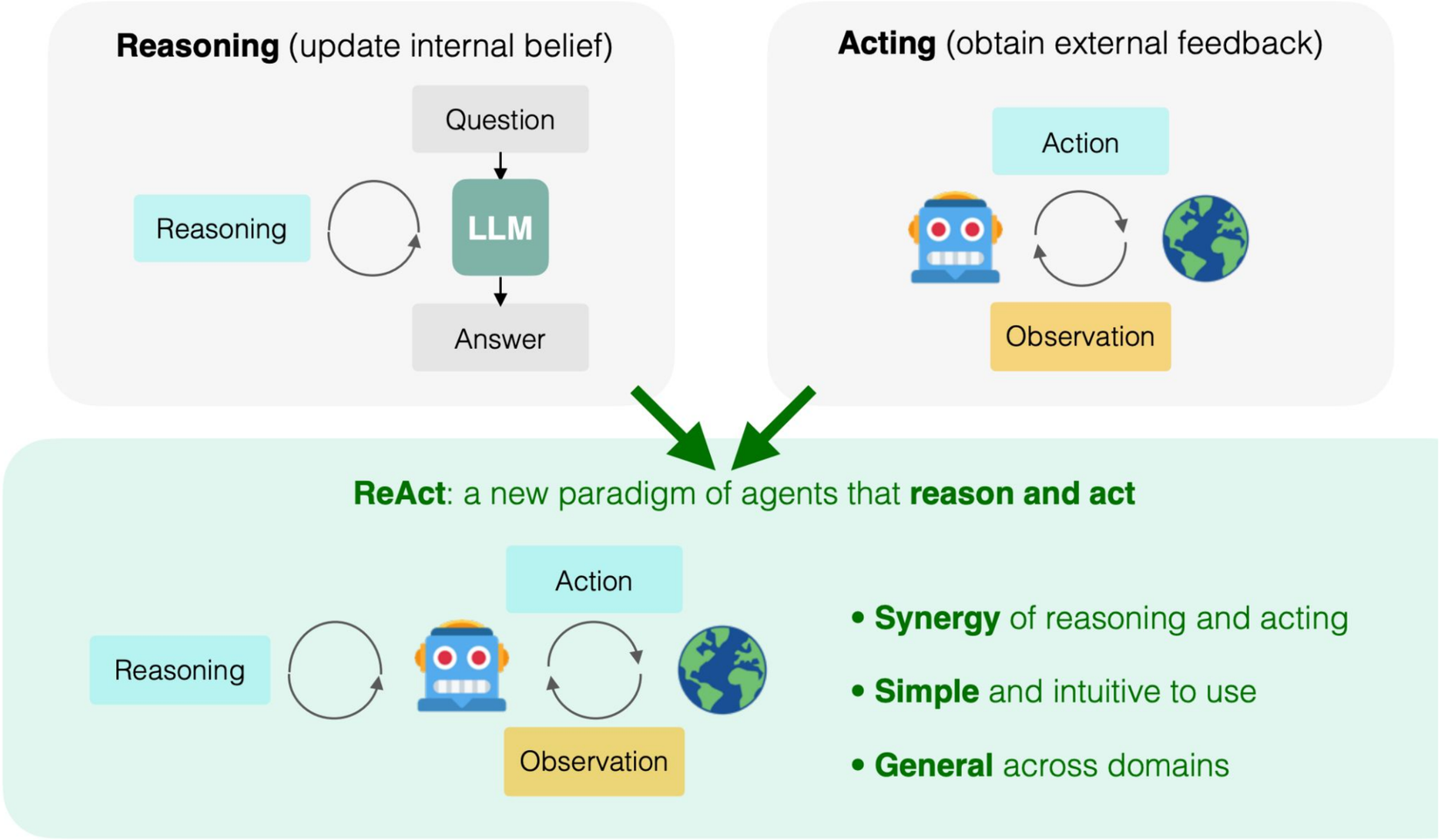


Retrieval
Search engine
Calculator
Weather API
Python
……

Lack of reasoning

Flexible and general to
augment knowledge,
computation, feedback, etc.

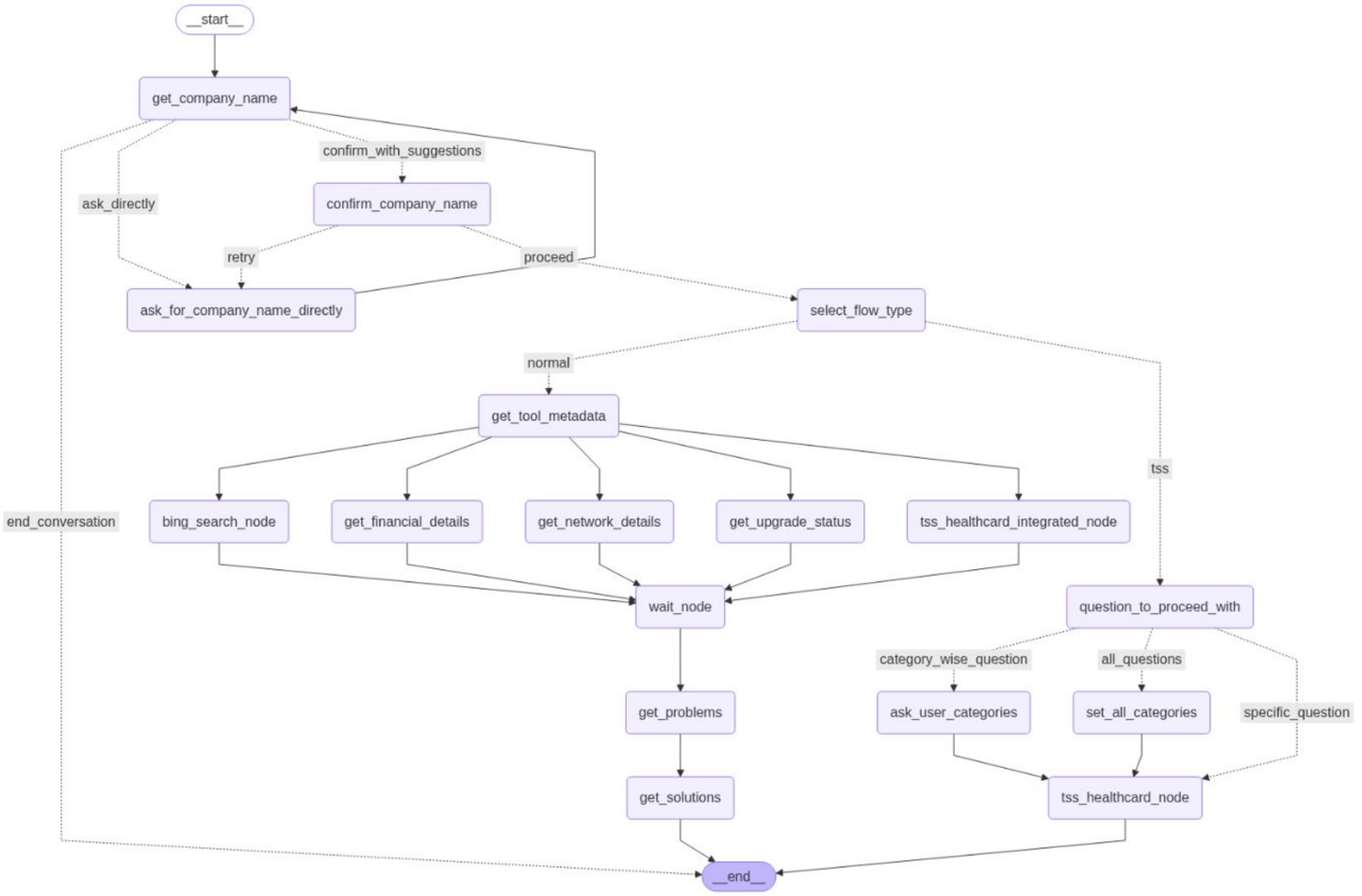
What's "LLM agent"?



Agentic Applications we created

1. **Customer Success Manager:** Multiple agents working together to create sales leads for a customer.
    a. V1 Released
    b. V2 WiP
2. **Case Handling (WiP):** Complete revamping of Nokia ticket management system.
3. **Gen BI:** Get insights from data by asking natural language questions
    a. PoC Success for Nokia client
    b. New Implementation for 3 Nokia internal stakeholders
        i. NDA
        ii. NPAS
        iii. NDPd

Agentic Applications we created



Customer Success Manager

Let’s do hands-on.

Step 1: Get Gemini API key (https://ai.google.dev/gemini-api/docs/api-key)

Step 2: Click on 

Step 3: Copy the key

Step 4: Let’s move to the notebook

https://colab.research.google.com/drive/150t9xorYg784TD7-nCAjXr-12EcrkbVS

https://bit.ly/ai-agent-talk

Sign up for my newsletter
We're hiring too!!
https://neuralreflections.com/



Google I/O Extended

https://tyagipy.bio.link/

Thank You

Twitter

LinkedIn